Chapter 6: Application of Derivatives

School Worksheet: step-wise solutions

20 questions **52** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ — 1 mark

A cylindrical tank is being filled with sand at the rate of $314\ \text{m}^3/\text{h}$. If the radius of the tank is 10 m, then the height of sand in the tank increases at the rate of: (Take $\pi = 3.14$)

(A) 1.1 m/h

(B) 1 **m/h**

(C) π m/h

(D) $\frac{\pi}{2}$ m/h

Formula / idea: Volume of a cylinder $V = \pi r^2 h$; here r is fixed, only h changes.

1 Write the volume with $r = 10$ m.
$V = \pi(10)^2 h = 100\pi h$

2 Differentiate w.r.t. t.
$\frac{dV}{dt} = 100\pi \frac{dh}{dt}$

3 Put $\frac{dV}{dt} = 314$ and $\pi = 3.14$. ✓ 1
$314 = 100(3.14)\frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{314}{314} = 1\ \text{m/h}$

4 Match with the options.
Option (B)

Answer **(B)** 1 **m/h**

Watch out: Treating r as a variable too. The tank's radius is fixed, so only h is differentiated.

2 MCQ 1 mark

The function $f(x) = x^3 - 3x^2 + 3x - 100$ is:

(A) **increasing on** $\mathbb{R}$ **(B)** decreasing on $\mathbb{R}$

(C) increasing on $(-\infty, 1)$ only **(D)** decreasing on $(1, \infty)$

Formula / idea: If $f'(x) \geq 0$ and $f'(x) = 0$ only at isolated points, f is strictly increasing.

1	Differentiate. $f'(x) = 3x^2 - 6x + 3$	
2	Factorise. $f'(x) = 3(x^2 - 2x + 1) = 3(x - 1)^2$	
3	A square is never negative. $f'(x) \geq 0$ for all $x \in \mathbb{R}$	
4	$f'(x) = 0$ only at $x = 1$, so f is increasing on $\mathbb{R}$. Option (A)	✓ 1

Answer **(A) increasing on** $\mathbb{R}$

Watch out: Choosing (C) because $f'(1) = 0$. One isolated zero of f' does not stop f increasing.

3 MCQ 1 mark

The interval in which $f(x) = x^2 - 4x + 6$ is strictly decreasing is:

(A) $(-\infty, 2)$ **(B)** $(2, \infty)$

(C) $(-\infty, \infty)$ **(D)** $(-2, 2)$

Formula / idea: f is strictly decreasing where $f'(x) < 0$.

1	Differentiate. $f'(x) = 2x - 4 = 2(x - 2)$	
2	Find where $f'(x) < 0$. $2(x - 2) < 0 \Rightarrow x < 2$	
3	So f is strictly decreasing on $(-\infty, 2)$. Option (A)	✓ 1

Answer **(A)** $(-\infty, 2)$

Watch out: Picking $(2, \infty)$: there $f'(x) > 0$, so f is increasing, not decreasing.

4 MCQ 1 mark

The maximum value of $\sin x \cos x$ is:

(A) 1 **(B)** $\frac{1}{2}$

(C) 2 **(D)** $\frac{1}{\sqrt{2}}$

Formula / idea: $\sin 2x = 2\sin x \cos x$ and $-1 \le \sin 2x \le 1$.

1 Use the double-angle formula.
$$\sin x \cos x = \frac{1}{2}\sin 2x$$

2 The greatest value of $\sin 2x$ is 1 (at $x = \frac{\pi}{4}$).
$$\frac{1}{2}\sin 2x \le \frac{1}{2}$$

3 So the maximum value is $\frac{1}{2}$. ✓ 1
Option (B)

Answer **(B)** $\frac{1}{2}$

Watch out: Answering 1 by taking the maximum of $\sin x$ and $\cos x$ separately; they are not both 1 at the same x.

5 MCQ 1 mark

The local minimum value of $f(x) = x + \frac{1}{x}$, $x > 0$, is:

(A) 1 **(B)** -2

(C) 2 **(D)** 0

Formula / idea: Second derivative test: $f'(c) = 0$ and $f''(c) > 0$ gives a local minimum at c.

1 Differentiate.
$$f'(x) = 1 - \frac{1}{x^2}$$

2 Put $f'(x) = 0$.
$$x^2 = 1 \Rightarrow x = 1 \quad (\because x > 0)$$

3 Second derivative at $x = 1$.
$$f''(x) = \frac{2}{x^3}, \quad f''(1) = 2 > 0$$

4 Local minimum at $x = 1$; find the value. ✓ 1
$f(1) = 1 + 1 = 2 \quad \Rightarrow$ Option (C)

Answer **(C)** 2

Watch out: Giving the point $x = 1$ (option A) instead of the value $f(1) = 2$.

6 **Assertion-Reason** 1 mark

Assertion (A): $f(x) = \sin x$ is strictly increasing in $\left(0, \frac{\pi}{2}\right)$. **Reason (R):** If $f'(x) > 0$ for every x in an open interval, then f is strictly increasing in that interval.

(A) **Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).**

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: Theorem: $f'(x) > 0$ on an open interval $\Rightarrow f$ is strictly increasing there.

1	Check A: $f'(x) = \cos x > 0$ for $0 < x < \frac{\pi}{2}$. So A is true. $f'(x) = \cos x > 0 \text{ in } \left(0, \frac{\pi}{2}\right)$	
2	Check R: it is the standard theorem in NCERT. So R is true.	
3	A follows directly from R using $f'(x) = \cos x > 0$. So R explains A.	
4	Hence the correct option is (A). Option (A)	✓ 1

Answer **(A) Both A and R are true, and R is the correct explanation of A.**

7 **Short answer** 2 marks

The edge of a cube is increasing at the rate of 3 cm/s. How fast is the volume increasing when the edge is 10 cm long?

Formula / idea: Rate of change: $\frac{dV}{dt} = \frac{dV}{dx} \cdot \frac{dx}{dt}$ (chain rule).

1	Let edge $= x$ cm and volume $= V$. Given rate: $V = x^3, \qquad \frac{dx}{dt} = 3 \text{ cm/s}$	
2	Differentiate w.r.t. t. $\frac{dV}{dt} = 3x^2 \frac{dx}{dt}$	✓ 1
3	Put $x = 10$ and $\frac{dx}{dt} = 3$. $\frac{dV}{dt} = 3(10)^2(3) = 900$	✓ ½

4 State the result with units. ✓ ½

$\text{Volume increases at } 900 \text{ cm}^3/\text{s}$

Answer **The volume is increasing at $900 \text{ cm}^3/\text{s}$.**

Watch out: Writing the unit as cm/s. The rate of volume is in cm^3/s.

Level 2

19 marks

8 **Short answer** 3 marks

Find the intervals in which $f(x) = 2x^3 - 3x^2 - 36x + 7$ is (a) strictly increasing (b) strictly decreasing.

Formula / idea: $f'(x) > 0 \Rightarrow$ strictly increasing; $f'(x) < 0 \Rightarrow$ strictly decreasing.

1 Differentiate.

$f'(x) = 6x^2 - 6x - 36$

2 Factorise. ✓ 1

$f'(x) = 6(x^2 - x - 6) = 6(x - 3)(x + 2)$

3 Put $f'(x) = 0$. ✓ ½

$x = -2 \quad \text{or} \quad x = 3$

4 These points split the real line. Check the sign of $f'(x)$ in each part. ✓ ½

Interval	Sign of $f'(x)$	Nature of f
$(-\infty, -2)$	$(-)(-) > 0$	strictly increasing
$(-2, 3)$	$(-)(+) < 0$	strictly decreasing
$(3, \infty)$	$(+)(+) > 0$	strictly increasing

Signs shown are of $(x - 3)$ and $(x + 2)$.

5 (a) Strictly increasing on ✓ ½

$(-\infty, -2) \cup (3, \infty)$

6 (b) Strictly decreasing on ✓ ½

$(-2, 3)$

Answer **(a) Strictly increasing on $(-\infty, -2) \cup (3, \infty)$; (b) strictly decreasing on $(-2, 3)$.**

Watch out: Factorising $x^2 - x - 6$ as $(x + 3)(x - 2)$. Check: $(x - 3)(x + 2) = x^2 - x - 6$.

9 **Short answer** 2 marks

Air is pumped into a spherical balloon at the rate of $900\ \text{cm}^3/\text{s}$. Find the rate at which the radius is increasing when the radius is 15 cm.

Formula / idea: Volume of a sphere $V = \frac{4}{3}\pi r^3$, so $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$.

1	Let radius $= r$ and volume $= V$. Given rate: $V = \frac{4}{3}\pi r^3, \qquad \frac{dV}{dt} = 900\ \text{cm}^3/\text{s}$	
2	Differentiate w.r.t. t. $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$	✓ 1
3	Put $r = 15$ and $\frac{dV}{dt} = 900$. $900 = 4\pi(15)^2 \frac{dr}{dt} = 900\pi \frac{dr}{dt}$	✓ ½
4	Solve for $\frac{dr}{dt}$. $\frac{dr}{dt} = \frac{900}{900\pi} = \frac{1}{\pi}\ \text{cm/s}$	✓ ½

Answer **The radius is increasing at $\frac{1}{\pi}$ cm/s.**

Watch out: Forgetting the chain rule factor $\frac{dr}{dt}$, or writing $3r^2$ instead of $4\pi r^2$.

10 **Short answer** 3 marks

Find the local maximum and local minimum values of $f(x) = x^3 - 6x^2 + 9x + 15$.

Formula / idea: Second derivative test: $f'(c) = 0$; $f''(c) < 0 \Rightarrow$ local max, $f''(c) > 0 \Rightarrow$ local min.

1	Differentiate and factorise. $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$	✓ ½
2	Put $f'(x) = 0$ for critical points. $x = 1 \quad \text{or} \quad x = 3$	✓ ½
3	Find the second derivative. $f''(x) = 6x - 12$	✓ ½
4	At $x = 1$: f'' is negative, so local maximum. $f''(1) = 6 - 12 = -6 < 0$	✓ ½
5	Local maximum value: $f(1) = 1 - 6 + 9 + 15 = 19$	✓ ½

6	At $x = 3$: f'' is positive, so local minimum. Find its value. $f''(3) = 6 > 0, \qquad f(3) = 27 - 54 + 27 + 15 = 15$	✓ ½
7	State both values. $\text{Local max value} = 19, \qquad \text{Local min value} = 15$	

Answer **Local maximum value** $= 19$ **(at** $x = 1$**); local minimum value** $= 15$ **(at** $x = 3$**).**

Watch out: Stopping at $x = 1$ and $x = 3$. The question asks for the values $f(1)$ and $f(3)$.

11 Short answer — 3 marks

Find the intervals in which the function $f(x) = \frac{\log x}{x}$ is strictly increasing or strictly decreasing.

Formula / idea: Quotient rule: $\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$; sign of $f'(x)$ decides increasing or decreasing.

1	Note the domain: $\log x$ needs $x > 0$. $\text{Domain} = (0, \infty)$	
2	Differentiate by the quotient rule. $f'(x) = \frac{x \cdot \frac{1}{x} - \log x \cdot 1}{x^2} = \frac{1 - \log x}{x^2}$	✓ 1
3	Put $f'(x) = 0$. $1 - \log x = 0 \;\Rightarrow\; \log x = 1 \;\Rightarrow\; x = e$	✓ ½
4	$x = e$ splits the domain into $(0, e)$ and (e, ∞). Note $x^2 > 0$ always. $\text{Sign of } f'(x) = \text{sign of } (1 - \log x)$	✓ ½
5	For $0 < x < e$: $\log x < 1$, so $f'(x) > 0$. f is strictly increasing on $(0, e)$	✓ ½
6	For $x > e$: $\log x > 1$, so $f'(x) < 0$. f is strictly decreasing on (e, ∞)	✓ ½

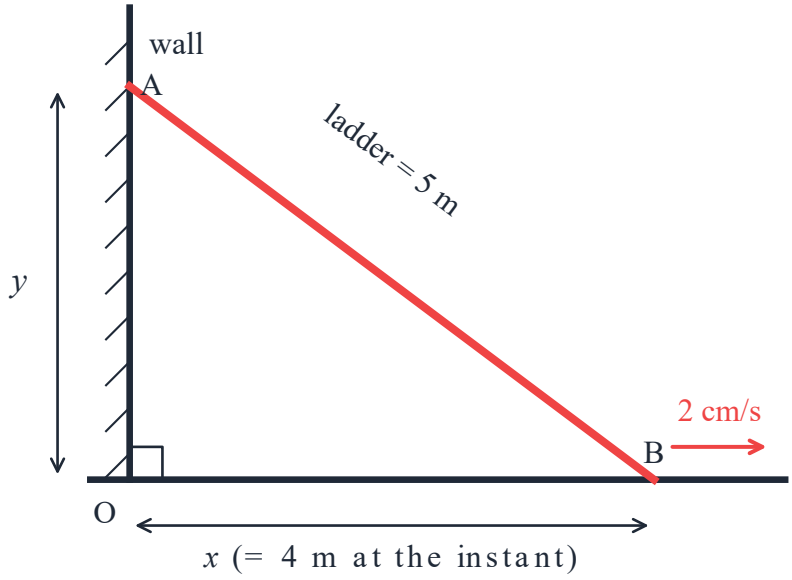
Answer **Strictly increasing on** $(0, e)$**; strictly decreasing on** (e, ∞)**.**

Watch out: Writing $(-\infty, e)$. The function is defined only for $x > 0$, so start from 0.

12 **Short answer** 3 marks

A ladder 5 m long leans against a wall. The bottom of the ladder is pulled along the ground away from the wall at 2 cm/s. How fast is its height on the wall decreasing when the foot of the ladder is 4 m from the wall?

Formula / idea: Pythagoras $x^2 + y^2 = 5^2$, then differentiate both sides w.r.t. t.



1	Let foot be x m from the wall and top y m high. By Pythagoras: $x^2 + y^2 = 25, \qquad \dfrac{dx}{dt} = 2 \text{ cm/s}$	✓ ½
2	Differentiate both sides w.r.t. t. $2x\dfrac{dx}{dt} + 2y\dfrac{dy}{dt} = 0 \Rightarrow \dfrac{dy}{dt} = -\dfrac{x}{y}\dfrac{dx}{dt}$	✓ 1
3	When $x = 4$, find y. $y = \sqrt{25 - 16} = 3$ m	✓ ½
4	Put $x = 4, y = 3, \dfrac{dx}{dt} = 2$. $\dfrac{dy}{dt} = -\dfrac{4}{3} \times 2 = -\dfrac{8}{3} \text{ cm/s}$ $\frac{x}{y}$ has no unit, so the answer stays in cm/s.	✓ ½
5	Negative sign means y is decreasing. State the result. Height decreases at $\dfrac{8}{3}$ cm/s	✓ ½

Answer **The height on the wall is decreasing at $\dfrac{8}{3}$ cm/s.**

Watch out: Answering $-\dfrac{8}{3}$ cm/s for "how fast is it decreasing". The rate of decrease is $\dfrac{8}{3}$ cm/s.

13 **Short answer** 3 marks

Find the absolute maximum and absolute minimum values of $f(x) = \sin x + \cos x$ on $[0, \pi]$.

Formula / idea: On a closed interval, compare f at the critical points and at both end points.

1	Differentiate. $f'(x) = \cos x - \sin x$	✓ ½
2	Put $f'(x) = 0$ and keep the root in $[0, \pi]$. $\tan x = 1 \;\Rightarrow\; x = \dfrac{\pi}{4}$	✓ ½
3	Value at the end point $x = 0$. $f(0) = 0 + 1 = 1$	✓ ½
4	Value at the critical point $x = \frac{\pi}{4}$. $f\left(\frac{\pi}{4}\right) = \dfrac{1}{\sqrt{2}} + \dfrac{1}{\sqrt{2}} = \sqrt{2}$	✓ ½
5	Value at the end point $x = \pi$. $f(\pi) = 0 + (-1) = -1$	✓ ½
6	Compare the three values. $\text{Abs. max} = \sqrt{2}$ at $x = \frac{\pi}{4}$, $\quad \text{Abs. min} = -1$ at $x = \pi$	✓ ½

Answer **Absolute maximum $= \sqrt{2}$ (at $x = \frac{\pi}{4}$); absolute minimum $= -1$ (at $x = \pi$).**

Watch out: Forgetting the end points. Here the minimum -1 occurs at the end point $x = \pi$.

14 **Short answer** 2 marks

Find two positive numbers whose sum is 15 and the sum of whose squares is minimum.

Formula / idea: Write S in one variable, put $S' = 0$, confirm minimum with $S'' > 0$.

1	Let the numbers be x and $15 - x$. Sum of squares: $S = x^2 + (15 - x)^2$	✓ ½
2	Differentiate and put $\dfrac{dS}{dx} = 0$. $\dfrac{dS}{dx} = 2x - 2(15 - x) = 4x - 30 = 0 \;\Rightarrow\; x = \dfrac{15}{2}$	✓ ½
3	Second derivative test. $\dfrac{d^2S}{dx^2} = 4 > 0 \;\Rightarrow\; S$ is minimum	✓ ½

4 Find the other number and conclude. ✓ ½

$15 - \frac{15}{2} = \frac{15}{2}$ Numbers: $\frac{15}{2}, \frac{15}{2}$

Answer **The numbers are $\frac{15}{2}$ and $\frac{15}{2}$.**

Watch out: Skipping $\frac{d^2S}{dx^2} > 0$. Without it you have not shown that S is a minimum.

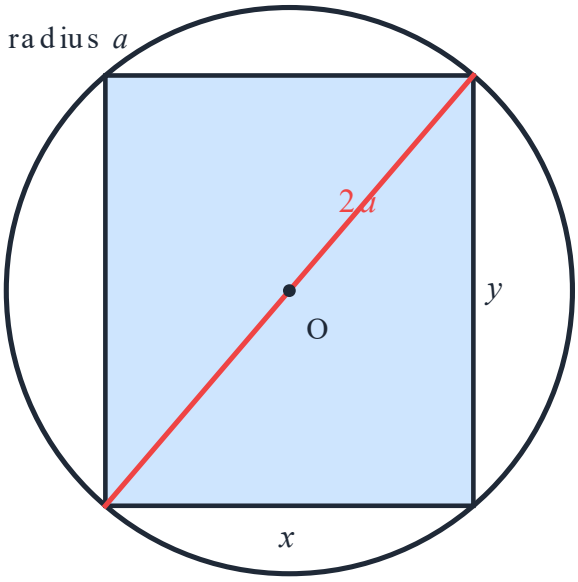
Level 3

25 marks

15 **Long answer** 5 marks

Show that of all rectangles inscribed in a given fixed circle, the square has the maximum area.

Formula / idea: Diagonal of an inscribed rectangle is a diameter: $x^2 + y^2 = (2a)^2$. Maximise A^2 instead of A.



1 Let the circle have radius a (fixed) and the rectangle have sides x, y.
$\text{Diagonal} = \text{diameter} = 2a$

2 By Pythagoras on the diagonal: ✓ 1
$x^2 + y^2 = 4a^2 \Rightarrow y = \sqrt{4a^2 - x^2}$

3 Write the area in terms of x. ✓ ½
$A = xy = x\sqrt{4a^2 - x^2}$

4 $A > 0$, so A is maximum when $S = A^2$ is maximum. ✓ ½
$S = A^2 = x^2(4a^2 - x^2) = 4a^2x^2 - x^4$

5	Differentiate S w.r.t. x. $\frac{dS}{dx} = 8a^2x - 4x^3 = 4x(2a^2 - x^2)$	✓ ½
6	Put $\frac{dS}{dx} = 0$ (take $x > 0$). $x^2 = 2a^2 \Rightarrow x = \sqrt{2}\,a$	✓ 1
7	Second derivative test at $x = \sqrt{2}\,a$. $\frac{d^2S}{dx^2} = 8a^2 - 12x^2 = 8a^2 - 24a^2 = -16a^2 < 0$	✓ 1
8	So S, hence A, is maximum at $x = \sqrt{2}\,a$. Find y. $y = \sqrt{4a^2 - 2a^2} = \sqrt{2}\,a = x$	✓ ½
9	Conclude. $x = y \Rightarrow$ the rectangle of maximum area is a square	

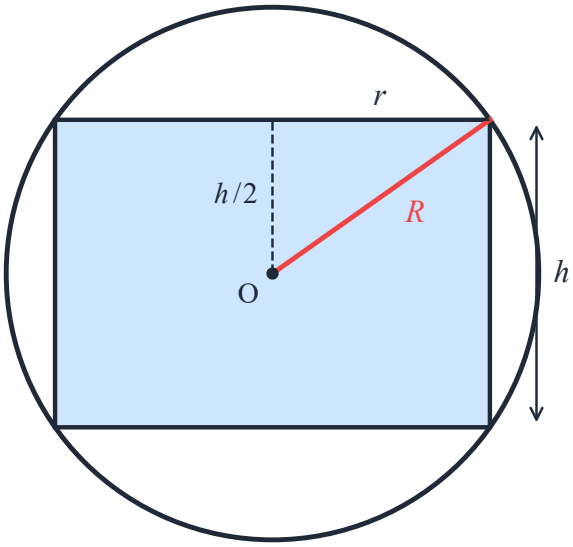
Answer **Area is maximum when $x = y = \sqrt{2}\,a$, i.e. the rectangle is a square.**

Watch out: Writing $x^2 + y^2 = a^2$. The diagonal is the diameter $2a$, not the radius a.

16 **Long answer** 5 marks

Show that the height of the cylinder of maximum volume that can be inscribed in a sphere of radius R is $\frac{2R}{\sqrt{3}}$. Also find the maximum volume.

Formula / idea: From the cross-section, $r^2 + \left(\frac{h}{2}\right)^2 = R^2$; volume of cylinder $V = \pi r^2 h$.



sphere of radius R (cross-section)

1	Let the cylinder have radius r and height h. In the right triangle from centre O: $r^2 + \left(\frac{h}{2}\right)^2 = R^2 \Rightarrow r^2 = R^2 - \frac{h^2}{4}$	✓ 1
2	Write the volume in terms of h only. $V = \pi r^2 h = \pi\left(R^2 h - \frac{h^3}{4}\right)$	✓ 1
3	Differentiate w.r.t. h. $\frac{dV}{dh} = \pi\left(R^2 - \frac{3h^2}{4}\right)$	✓ ½
4	Put $\frac{dV}{dh} = 0$. $h^2 = \frac{4R^2}{3} \Rightarrow h = \frac{2R}{\sqrt{3}}$	✓ 1
5	Second derivative test. $\frac{d^2V}{dh^2} = -\frac{3\pi h}{2} < 0$ at $h = \frac{2R}{\sqrt{3}}$ $h > 0$, so $-\frac{3\pi h}{2}$ is negative.	✓ ½
6	So V is maximum when $h = \frac{2R}{\sqrt{3}}$. Hence shown.	

7 Find r^2 at this height. ✓ ½

$$r^2 = R^2 - \frac{1}{4} \cdot \frac{4R^2}{3} = \frac{2R^2}{3}$$

8 Maximum volume: ✓ ½

$$V_{\max} = \pi \cdot \frac{2R^2}{3} \cdot \frac{2R}{\sqrt{3}} = \frac{4\pi R^3}{3\sqrt{3}} \text{ cubic units}$$

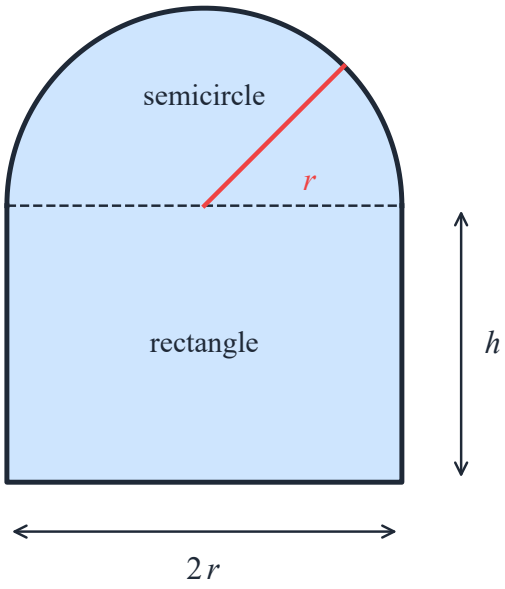
Answer **Height** $= \frac{2R}{\sqrt{3}}$**; maximum volume** $= \frac{4\pi R^3}{3\sqrt{3}}$ **cubic units.**

Watch out: Using $r^2 + h^2 = R^2$. Only half the height, $\frac{h}{2}$, lies between the centre and the top face.

17 **Long answer** 5 marks

A window is in the form of a rectangle surmounted by a semicircular opening. The total perimeter of the window is 10 m. Find the dimensions of the window so as to admit maximum light through the whole opening.

Formula / idea: Maximum light means maximum area. Perimeter = base + two sides + semicircular arc πr.



1 Let radius of semicircle $= r$, so width $= 2r$; height of rectangle $= h$. ✓ 1

$$\text{Perimeter: } 2r + 2h + \pi r = 10$$

2 Express h in terms of r.

$$h = \frac{10 - 2r - \pi r}{2}$$

3 Area of the window = rectangle + semicircle. ✓ ½

$$A = 2rh + \frac{1}{2}\pi r^2$$

4 Substitute h and simplify. ✓ ½

$$A = r(10 - 2r - \pi r) + \frac{\pi r^2}{2}$$
$$= 10r - 2r^2 - \frac{\pi r^2}{2}$$

5 Differentiate w.r.t. r. ✓ ½

$$\frac{dA}{dr} = 10 - 4r - \pi r$$

6 Put $\frac{dA}{dr} = 0$. ✓ 1

$$r(\pi + 4) = 10 \;\Rightarrow\; r = \frac{10}{\pi + 4}$$

7 Second derivative test. ✓ ½

$$\frac{d^2A}{dr^2} = -(4 + \pi) < 0 \;\Rightarrow\; A \text{ is maximum}$$

8 Width of the window: ✓ ½

$$2r = \frac{20}{\pi + 4}\ \text{m}$$

9 Height of the rectangle: put $r = \frac{10}{\pi + 4}$ in h. ✓ ½

$$h = \frac{1}{2}\left[10 - \frac{10(\pi + 2)}{\pi + 4}\right] = \frac{1}{2} \cdot \frac{20}{\pi + 4} = \frac{10}{\pi + 4}\ \text{m}$$

Answer **Width $= \frac{20}{\pi + 4}$ m, height of rectangle $= \frac{10}{\pi + 4}$ m (radius of semicircle $= \frac{10}{\pi + 4}$ m).**

Watch out: Adding the top side $2r$ to the perimeter. The top of the rectangle is covered by the semicircle, so it is not on the boundary.

18 Short answer — 3 marks

Find the intervals in which $f(x) = \sin x + \cos x$, $0 \le x \le 2\pi$, is strictly increasing or strictly decreasing.

Formula / idea: $f'(x) > 0 \Rightarrow$ strictly increasing; $f'(x) < 0 \Rightarrow$ strictly decreasing.

1 Differentiate. ✓ ½

$$f'(x) = \cos x - \sin x$$

2 Put $f'(x) = 0$ and find roots in $[0, 2\pi]$. ✓ ½

$$\tan x = 1 \;\Rightarrow\; x = \frac{\pi}{4},\ \frac{5\pi}{4}$$

3 Check the sign of $f'(x)$ in each interval using a test point. ✓ 1

Interval	Test point: $f'(x)$	Nature of f
$\left[0, \frac{\pi}{4}\right)$	$f'(0) = 1 > 0$	strictly increasing
$\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$	$f'\left(\frac{\pi}{2}\right) = -1 < 0$	strictly decreasing
$\left(\frac{5\pi}{4}, 2\pi\right]$	$f'\left(\frac{3\pi}{2}\right) = 1 > 0$	strictly increasing

4 Strictly increasing on
$\left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$ ✓ ½

5 Strictly decreasing on
$\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ ✓ ½

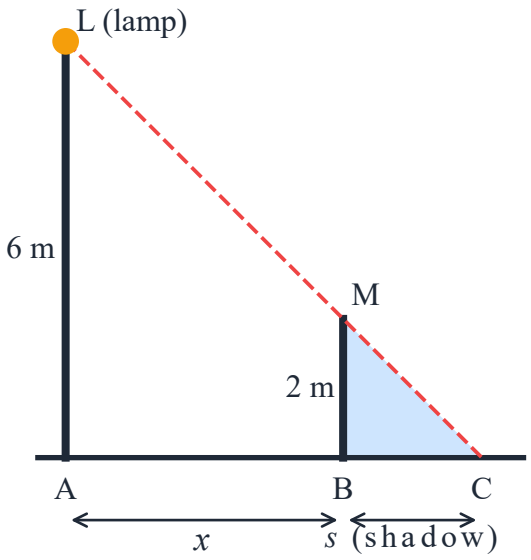
Answer **Strictly increasing on $\left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$; strictly decreasing on $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$.**

Watch out: Missing the root $x = \frac{5\pi}{4}$. In $[0, 2\pi]$, $\tan x = 1$ has two roots.

19 **Short answer** 3 marks

A man 2 m tall walks at a uniform speed of 5 km/h away from a lamp post 6 m high. Find the rate at which the length of his shadow increases.

Formula / idea: Similar triangles link the shadow s to the distance x; then differentiate w.r.t. t.



1 Let AB $= x$ m (man from post), BC $= s$ m (shadow).
$\frac{dx}{dt} = 5$ km/h

2 Triangles CBM and CAL are similar (lamp L, man's head M). ✓ 1
$\frac{AL}{BM} = \frac{AC}{BC} \Rightarrow \frac{6}{2} = \frac{x+s}{s}$

3 Simplify. ✓ ½
$3s = x + s \Rightarrow s = \frac{x}{2}$

4 Differentiate w.r.t. t. ✓ ½
$\frac{ds}{dt} = \frac{1}{2}\frac{dx}{dt}$

5	Put $\frac{dx}{dt} = 5$. $\frac{ds}{dt} = \frac{1}{2} \times 5 = \frac{5}{2}\ \text{km/h}$	✓ ½
6	State the result. $\text{Shadow increases at } \frac{5}{2}\ \text{km/h}$	✓ ½

Answer **The shadow increases at $\frac{5}{2}$ km/h.**

Watch out: Taking the shadow's tip speed ($\frac{15}{2}$ km/h) as the answer. The question asks for the rate of the shadow's length.

20 **Case study** 4 marks

Case Study: A technical company is designing a rectangular solar panel installation on a roof using 300 metres of boundary material. The design includes a partition running parallel to one of the sides, dividing the area into two sections. Let the length of the side perpendicular to the partition be x metres and the side parallel to the partition be y metres.

(i) Write the equation for the total boundary material used (boundary and partition) in terms of x and y. 1 mark

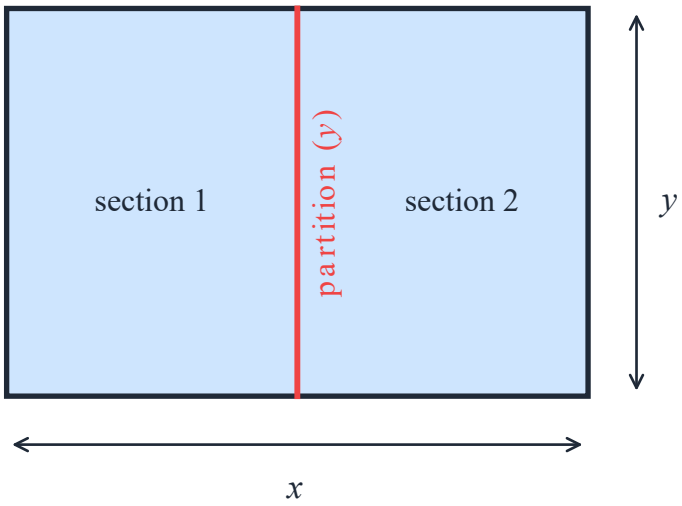
(ii) Write the area of the solar panel as a function of x. 1 mark

(iii)(a) Find the critical point of the area function and use the second derivative test to find the maximum area. 2 marks

OR

(iii)(b) Using the first derivative test, find the maximum area the company can enclose with the 300 m of boundary material. 2 marks

Formula / idea: Use the material constraint to write area in one variable, then apply the derivative test.



(i) 1 mark

1 The boundary has two sides of length x and two of length y.
$\text{Boundary} = 2x + 2y$ ✓ ½

2 The partition is parallel to the y sides, so its length is y. Total:
$2x + 2y + y = 300 \Rightarrow 2x + 3y = 300$ ✓ ½

Answer (i) $2x + 3y = 300$

(ii) 1 mark

1 From (i), express y in terms of x.
$y = \dfrac{300 - 2x}{3}$ ✓ ½

2 Area $= xy$.

$$A(x) = x \cdot \frac{300 - 2x}{3} = 100x - \frac{2x^2}{3}, \quad 0 < x < 150$$

✓ ½

Answer (ii) $A(x) = 100x - \dfrac{2x^2}{3}$

(iii)(a)

2 marks

1 Differentiate $A(x)$.

$$A'(x) = 100 - \frac{4x}{3}$$

✓ ½

2 Put $A'(x) = 0$ for the critical point.

$$\frac{4x}{3} = 100 \;\Rightarrow\; x = 75$$

✓ ½

3 Second derivative test.

$$A''(x) = -\frac{4}{3} < 0 \;\Rightarrow\; A \text{ is maximum at } x = 75$$

✓ ½

4 Find y and the maximum area.

$$y = \frac{300 - 150}{3} = 50, \qquad A = 75 \times 50 = 3750 \text{ m}^2$$

✓ ½

Answer (iii)(a) **Critical point $x = 75$; maximum area $= 3750$ m^2.**

OR

(iii)(b)

2 marks

1 Differentiate $A(x)$.

$$A'(x) = 100 - \frac{4x}{3}$$

✓ ½

2 Put $A'(x) = 0$.

$$x = 75$$

✓ ½

3 Sign of A' near 75: positive for $x < 75$, negative for $x > 75$.

$$A'(74) = \frac{4}{3} > 0, \qquad A'(76) = -\frac{4}{3} < 0$$

A' changes from $+$ to $-$, so local maximum.

✓ ½

4 So A is maximum at $x = 75$, $y = 50$.

$$A_{\max} = 75 \times 50 = 3750 \text{ m}^2$$

✓ ½

Answer (iii)(b) **Maximum area $= 3750$ m^2 (at $x = 75$ m, $y = 50$ m).**

Watch out: Writing $2x + 2y = 300$ and forgetting the partition, which uses another y metres of material.